A Multimodal Approach for Early Identification of Mild Cognitive Impairment and Alzheimer's Disease With Fusion Network Using Eye Movements and Speech

Hasnain Ali Shah, Sami Andberg, Anne M. Koivisto, and Roman Bednarik

***Abstract*— Detecting Alzheimer's disease (AD) in its earliest stages, particularly during an onset of Mild Cognitive Impairment (MCI), remains challenging due to the overlap of initial symptoms with normal aging processes. Given that no cure exists and current medications only slow the disease's progression, early identification of at-risk individuals is crucial. The combination of eye-tracking and speech analysis offers a promising diagnostic solution by providing a non-invasive method to examine differences between healthy controls and individuals with MCI, who may progress to develop AD. In this study, we analyzed a multimodal clinical eye-tracking and speech dataset collected from 78 participants (37 controls, 20 MCI, and 21 AD) during the King-Devick test and a reading task to classify and diagnose MCI/AD versus healthy controls. To that end, we developed a Fusion Neural Network, a deep learning-based classification model that integrates gaze and speech-derived features, including pupil size variations, fixation duration, saccadic movements, and speech delay, to improve MCI diagnosis performance. We achieved an average classification accuracy of 79.2% for MCI diagnosis and 82% for AD. Our findings indicate that features related to pupil size and eye-speech temporal dynamics are strong indicators for detection tasks. Moreover, the results indicate that using multimodal data (gaze + speech) significantly improves classification accuracy compared to unimodal data from speech or gaze alone.**

***Index Terms*— Alzheimer's disease, deep learning, dementia, eye movements, eye-tracking, mild cognitive impairment, speech.**

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I. Introduction

DEMENTIA affects approximately 55 million people worldwide, with nearly 10 million new cases every year, and is considered one of the most expensive diseases [1]. Alzheimer's disease (AD) accounts for 60-70% of dementia cases, leading to a substantial percentage of cases globally [2]. AD is a neurological disorder primarily affecting memory, reasoning, and judgment. It is defined by a steady decline in cognitive skills. At the cellular level, it is characterized by the accumulation of tau tangles and beta-amyloid plaques in the brain [3]. Its insidious nature, marked by a decline in memory, reasoning, and cognitive abilities, poses significant challenges to affected individuals and has repercussions that reach beyond the individuals who are affected, including families, caregivers, and healthcare systems.

Mild cognitive impairment (MCI) is defined as a precursor to AD. It is a transitional stage between the normal cognitive decline associated with aging and the more severe cognitive decline observed in dementia. Individuals suffering from MCI/AD initially complain of memory dysfunctions, where cognitive issues begin to impact day-to-day functioning and gradually progress into a completely demented state [4]. Early identification of MCI/AD is challenging due to the complex nature and spectra of MCI symptoms that frequently resemble age-related cognitive decline. Standard cognitive tests during clinical examinations usually fail to detect MCI, mainly if it is in the initial stages [5]. Therefore, addressing the detection difficulty is vital as it provides a window of opportunity for just-in-time treatment that can delay or prevent the progression to AD [6], [7].

Clinicians and health experts utilize a wide range of advanced diagnostic technologies such as Magnetic Resonance Imaging, blood analysis, and cognitive tests to identify specific markers or behavioral changes indicative of MCI and AD [8]. Recently, eye-tracking (ET) technology has emerged as a non-invasive approach with significant potential for early detection of AD [9], [10], [11]. This method involves tracking and analyzing eye movement patterns, including variations in gaze patterns, pupillary responses, and gaze trajectories, which

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can hint at neurological deviations [12], [13]. A study by Boz et al. [14] found that AD patients had shorter and smaller fixations on targets. Similarly, pupil response variations have been studied to understand emotional states and cognitive effort differences in MCI patients [15], [16].

Eye movement abnormalities can be detected using various experimental tasks. One such approach is the saccade/antisaccade task, where participants are required to either fixate on or look away from a presented stimulus. This method has shown promising results in identifying MCI and AD-related impairments [17], [18], [19], [20], [21], [22]. Notably, Fernández et al. [23] demonstrated that eye-tracking differences are detectable even in cognitively normal individuals with a family history of late-onset AD, suggesting that biomarkers may emerge before cognitive symptoms become apparent. One widely used eye-tracking-based cognitive assessment tool is the King-Devick (KD) Test, which evaluates rapid eye movements and visual processing speed by requiring participants to read aloud sequences of numbers as quickly and accurately as possible [10]. Galetta et al. [24] utilized the KD test to effectively distinguish between cognitively normal individuals and those with MCI or AD, achieving an area-under-curve (AUC) score of 0.72–0.74. A detailed description of the KD test is provided in Section II.

Research has shown that language and speech analysis play a significant role in detecting cognitive impairments. Language is one of the first domains affected by AD, and longitudinal studies have demonstrated that subtle linguistic alterations can appear a year or more before a formal dementia diagnosis [25]. Recent studies increasingly emphasize the advantages of integrating language and eye movement data to enhance MCI/AD classification accuracy. Jang et al. [26] introduced innovative tasks, such as memory description and pupil calibration, analyzing eye movements alongside speech features. Their approach achieved an impressive overall AUC of 0.83 $\pm$ 0.01. Similarly, Barral et al. [27] utilized a multimodal dataset incorporating eye-tracking and speech data during the Cookie Theft task, attaining 80% accuracy in distinguishing AD patients from healthy controls.

The integration of machine learning (ML) and deep learning (DL) with eye-tracking and speech technology has emerged as a powerful tool for identifying Alzheimer's disease and its stages [28]. While researchers have explored eye-tracking and speech separately [29], only a few studies have investigated their combined potential [30]. Sriram et al. [31] developed a lightweight DL-based VTNet classifier for Alzheimer's classification, demonstrating improvements over the results obtained by Jang et al. [26]. In another study, Chen et al. [6] introduced ERPNet, a feature extraction layer designed to capture multimodal features from eye movements and EEG data, achieving an accuracy of 88.81% for MCI diagnosis and 100% for AD diagnosis.

This study expands on our earlier journal paper [10], where we highlighted the potential of eye movement data from the KD test. Expanding on that work, we introduce a multimodal approach integrating eye-tracking during a spontaneous speech task analysis. The major contributions of this work are as follows:

TABLE I

DEMOGRAPHIC BREAKDOWN OF THE TEST COHORT, ORIGINALLY PUBLISHED IN [10]. VALUES ARE PRESENTED AS MEANS (±SD) OR PERCENTAGES (N). CO, CONTROL; MCI, PERSONS WITH MILD COGNITIVE IMPAIRMENT; AD, ALZHEIMER'S DISEASE; APOE4, APOLIPOPROTEIN E4; SD, STANDARD DEVIATION. TOTAL N = 19. BOLDED *p*-VALUES INDICATE SIGNIFICANT DIFFERENCES BETWEEN GROUPS

Category	CO	MCI	AD	*p*
No of participants	37	20	21	
Gender, female (N)	54 (20)	45 (9)	61.9 (13)	0.555
Age, years (SD)	71 (5.1)	72 (6.4)	71 (6.8)	0.961
Education, years (SD)	13 (4.3)	11 (3.7)	13 (4.3)	0,278
Family history (N)	51 (19)	45 (9)	57 (12)	0.739
APOE4 carrier (N)	37.8 (14)	35 (7)	78.9 (15)*	**0.006**
APOE4 two alleles (N)	0 (0)	10 (2)	32 (6)*	**0.001**

- We analyzed multimodal data comprising eye movements and speech from 78 participants (37 controls, 20 MCI, and 21 AD) using both the KD test and a newly introduced reading task to investigate key features related to MCI/AD that the KD test might have overlooked.
- We developed a DL-based algorithm named Fusion Neural Network (FNN), which incorporates eye movement features, speech features, and pupillary features to perform MCI, AD, and control classification. Our findings demonstrated that the FNN achieved an AUC of 0.82 on the reading tasks and 0.79 on KD, effectively classifying the patients into their respective categories.
- We extracted various feature sets from our multimodal clinical eye-tracking dataset using various scales and focus features and analyzed the results with Shapley Additive explanations (SHAP) to establish which features were most beneficial for correctly classifying the individuals at risk.

II. METHODS AND MATERIALS

A. Cohort

In this study, we utilized the multimodal BEGAD study dataset, originally evaluated statistically in our previous study [10]. The test cohort consists of 78 participants (37 controls, 20 MCI, and 21 AD), the same cohort used in our prior work, as reported in Table I. The persons classified as MCI after the cognitive evaluation were initially taken in as controls and had their diagnosis changed to MCI after the cognitive evaluation. The Alzheimer's group was only mildly demented (MMSE 23.9). The CERAD-NB and MMSE assessments were utilized to evaluate cognitive function. The study was conducted in accordance with the principles of the Declaration of Helsinki and received approval from the Research Ethics Committee of the Northern Savo Hospital District (Approval Number: Dnro 482/2017). All participants gave informed consent prior to their involvement in the study. They also completed the Boston Naming test, the clock drawing test, and the constructional praxis delayed recall test recommended by guidelines to help health professionals in diagnosing MCI and AD. A future expansion of the dataset with additional participants is planned, but this study presents results based solely on the current cohort.

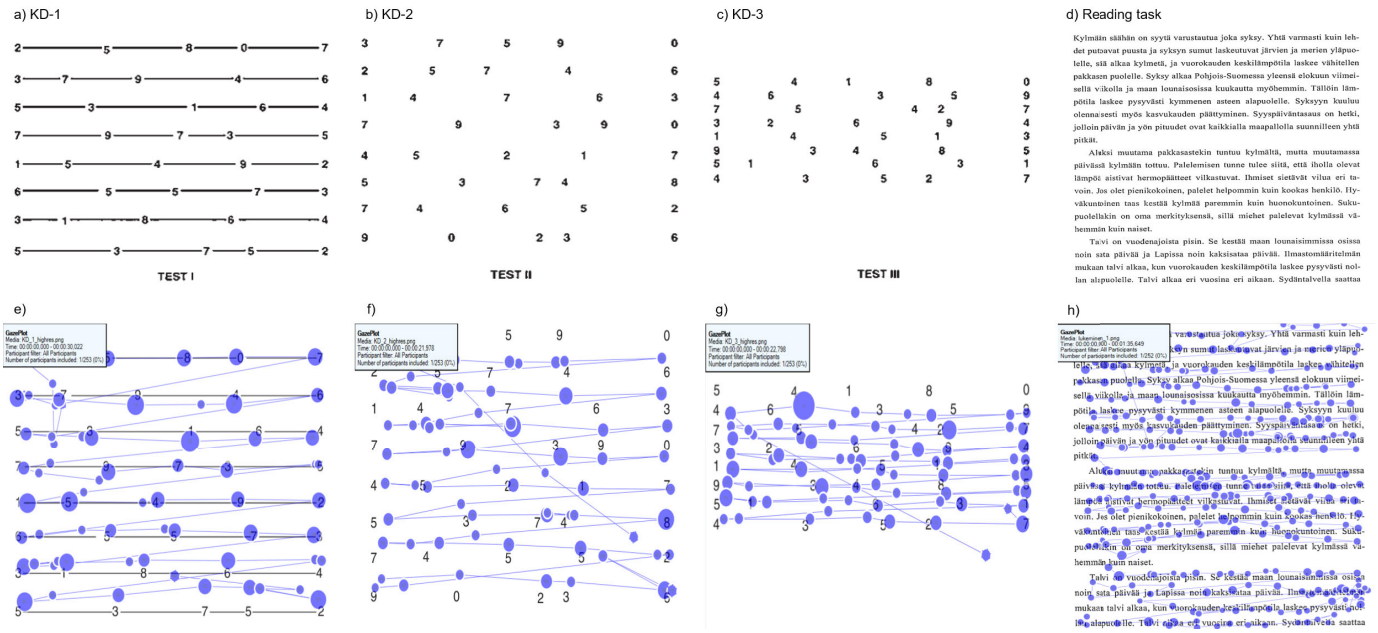


Fig. 1. Illustration of the King-Devick test cards 1, 2, 3 (a, b, c) and the reading task (d), alongside the corresponding eye-tracking data of a control participant (e, f, g, h) for the same tasks. The circular markers represent fixations, while the connecting lines depict saccades, providing insights into gaze behavior and task engagement patterns.

B. King-Devick Test

The King-Devick test is a standardized, commercial reading test that has been used to evaluate concussions [32], which has also been utilized in relation to detecting differences between the cognitively healthy and people with Alzheimer's disease [10], [24], [33]. In the KD test, the participants read aloud each number on the screen from left to right, row by row, with both eyes open, presented in Figure 1a. Each test card consists of 40 digits arranged in five rows, with variable spacing between the numbers both within and across rows. The visual challenge of the test cards increases progressively. The first test card features straight lines connecting the numbers to facilitate visual scanning. The second test card removes these connecting lines. The final test card not only lacks connecting lines but also has reduced spacing between the rows, heightening the visual difficulty. The King-Devick scoring includes the total time used for the three cards combined and the total number of errors made.

C. Reading Task

For the reading part, we used a standardized text from the Niilo Mäki Institute [34]. The text, presented in Figure 1b, is in Finnish, which is the native language of the test cohort. It describes the colder weather in autumn and the need for additional clothing to stay comfortable outside. The text comprises 159 words, including both familiar and some less common ones, such as 'ilmastomääritelmän,' which roughly translates to the genitive form of a 'definition of climate' in English. This word, among others, may not have been previously spoken aloud by the participants, making these Regions of Interest (ROIs) excellent for focused analysis. The objective of the reading task is to identify common deficits associated with AD, such as reduced reading speed and increased word fixations or re-fixations.

D. Eye-Tracking Recordings and Apparatus

The eye-racking recordings were conducted at a local clinic using an integrated Tobii TX300 unit, which includes a monitor and a 300Hz eye tracker from Tobii Technology AB, Sweden. The recordings took place in an office room with fluorescent lighting and occasional daylight. Participants were seated 60 cm away from the screen, and the eye-racking was calibrated using the standard 9-point calibration at the medium speed preset. A headrest was not used because participants needed to speak during the test. The internal movement compensation of the Tobii unit was deemed sufficient to manage small head movements without compromising data quality. Recordings, including eye-tracking data and audio and video feeds of the participants, were made using Tobii Studio 3.2.1 software [35].

1) Quality Control: There were multiple subjects where eye-tracking data for a part of the screen was lost. This was often due to drooping eyelids or strong multifocal glasses that caused the test subjects to tilt their head upwards when reading the top part of the screen, thus losing the eye-tracking a part of the screen where the signal was not reliable. We decided that persons with less than 60% success rate were removed from the analysis (4 in control, 1 in MCI, 4 in AD groups). Some of the audio recordings for some subjects were of low quality, making their data unsuitable for automatic audio processing speech analysis (4 in control, 2 in MCI, 1 in AD). The eye-tracking recordings were carefully quality-controlled and exported using Tobii Studio 3.4.8 software. The export utilized the built-in I-VT filter with the recommended values: 75ms max gap length interpolation, 20 ms window length, 30 deg/s

TABLE II
ROI-BASED FEATURES EXTRACTED FROM THE PROCESSED DATA FOR MACHINE LEARNING ANALYSIS. LAST FIXATION REFERS TO THE FIXATION AT THE END OF THE LAST SACCADE THAT BEGAN FROM THE ROI AREA

Category	Feature	Description
Visual Features	ROI	The number of ROI being evaluated
	Number of fixations	Number of fixations hit on a ROI
	No of saccades	Number of saccades starting or ending on a ROI
	No of short saccades (<1 ang)	Number of shorter saccades starting or ending on a ROI
	No of long saccades (>2 ang)	Number of longer saccades starting or ending on a ROI
	N of regressive sac	Number of regressive saccades (gaze direction left) on a ROI
	Saccade amplitude sum	Sum of the amplitudes of all saccades on a ROI
	Total fixation time	Total fixation time on a ROI
Pupillary Features	Left pupil at 1st fix	Left pupil size at the first fixation on a ROI
	Left pupil at last fix	Left pupil size at the last fixation on an ROI
	Left pupil difference	Difference in left pupil size between first and last fixation on a ROI
	Right pupil at 1st fix	Right pupil size at the first fixation on an ROI
	Right pupil at last fix	Right pupil size at the last fixation on an ROI
	Right pupil difference	Difference in right pupil size between first and last fixation on a ROI
	Min left pupil	Minimum left pupil size during the fixation on an ROI
	Max left pupil	Maximum left pupil size during the fixation on an ROI
	Min-Max difference	Difference between minimum and maximum left pupil size on an ROI
Speech Features	Total spoken time	Total time spent speaking out loud on a ROI
	Delay of 1st fix to utterance (in sec)	Delay in seconds from the first fixation on an ROI to the beginning of saying the ROI
	Delay of last fix to end of utterance (in sec)	Delay in seconds from the last fixation on an ROI to finishing saying the ROI

velocity threshold, merging adjacent fixations between 75ms and 0.5deg, and discarding fixations shorter than 60ms [36].

2) Eye-Tracking Data Pre-Processing: The exported data was further processed using custom-made Python scripts, which had information about the ROIs related to each task. A pupil size baseline was calculated for each participant using an average value of the first 10 successful fixation results from the first instruction slide shown to participants after the calibration. The pupil-based metrics presented in the features have this baseline subtracted from them, thus measuring the difference in pupil size at a given time from the initial condition at the start of the testing.

As Tobii Studio export cuts out every fixation outside of the displayed media box, these values were read from the other unadjusted export fields, adjusted to the media window, and inserted back into the data. The resulting new gaze data was then compared to the known ROI areas of each task at the top and bottom rows, and left and right edges of ROIs on the media. In case the subject's gaze points seemed to have been globally shifted in relation to the ROIs, a four-way recalibration field was calculated (one value for each edge). The gaze point data was then recalibrated for each task using these task-specific values and the distances of each gaze point from the edges. This noticeably improved the ROI hit accuracy on many recordings, even if the recalibration field was a task-relevant static box. Further improvements for the data accuracy of a few special subjects could be achieved if the recalibration field could be also skewed when needed, but this was not implemented at this point.

E. Speech Data Pre-Processing

The successful audio recordings were normalized using Audacity 2.3.3 and all utterances that were not part of the expected test output were removed by muting them [37]. The speech in the audio files was then time-synchronized to a list of the expected spoken words using Aalto-ASR-align from AaltoASR 1.1 module on CSC's Taito supercluster [38]. The resulting TextGrid file had the timings of the ROI words, when applicable. These TextGrids were then manually evaluated using ELAN 5.2 towards the original audio files, and all speech that was previously out was transliterated by hand. At the same time, also any mistakes, misspoken words, extra words, missing words, and repetitions were tagged. Manual timing adjustments to transliteration were made in ELAN when necessary [39].

F. Feature Extraction

The preprocessed eye-tracking data, audio data, and Tobii's internal metadata, accessed from the SQLite database file in the Tobii project folder, was further processed using custom-made Python stack. We developed a stack for ingesting data from the three sources and combining them for feature analysis purposes into a single stream. For the ROI analysis we focused on the specific literals, numbers and words (ROIs) the participants had to read out loud. The following features were extracted for each ROI as presented in Table II. In this analysis we focused on the specific ROIs, meaning these regions were either the bounding boxes of the numbers in the King-Devick tasks, or the words in the reading task.

G. Fusion Neural Network

In this study, we developed a simple yet effective DL-based Fusion Neural Network to classify participants into MCI, AD, and control groups. The architecture leverages modality-specific neural networks to extract features from visual, speech, and pupillary data, subsequently fused to enhance classification performance. Before delving into the implementation details, let us mathematically formulate the model's components. The dataset used in this study is denoted as $D = \left\{\left(X_v^i, X_p^i, X_s^i, y^i\right)\right\}_{i=1}^{N}$, where N represents the total number of participants included in the cohort. For each participant i, $X_v^i \in \mathbb{R}^{n_v \times d_v}$ corresponds to the visual features, where n_v is the number of time steps or samples

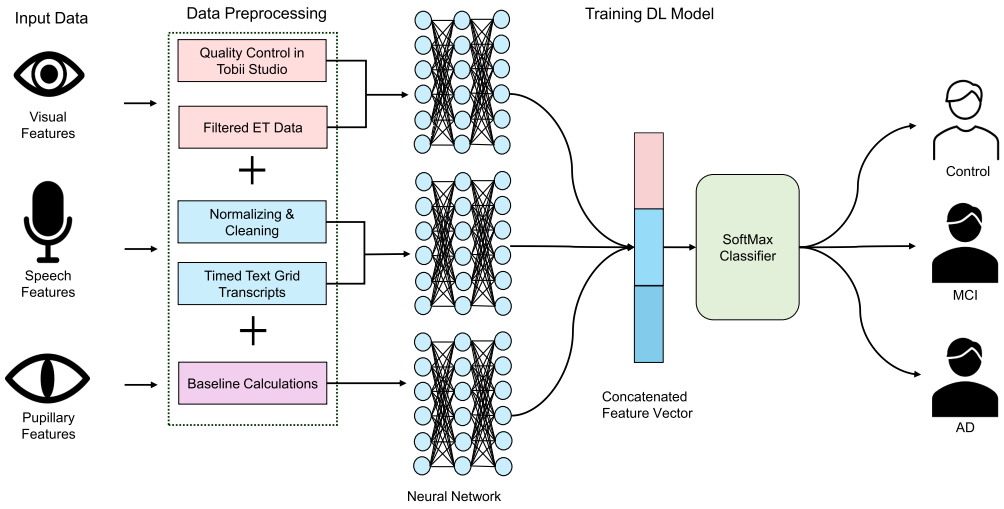


Fig. 2. Flowchart of data preprocessing and the training process.

and d_v is the dimensionality of the visual feature vector. Similarly, $X_p^i \in \mathbb{R}^{n_p \times d_p}$ represents the pupillary features, with n_p denoting the number of time steps or samples and d_p indicating the dimensionality of the pupillary feature vector. The speech features are captured as $X_s^i \in \mathbb{R}^{n_s \times d_s}$, where n_s is the number of time steps or samples, and d_s is the dimensionality of the speech feature vector. Each participant is also associated with a class label $y^i \in \{0, 1, 2\}$, where 0 corresponds to healthy controls, 1 indicates individuals with MCI, and 2 represents individuals with AD. This dataset structure ensures that multimodal data from visual, pupillary, and speech features can be systematically utilized to predict the corresponding cognitive status of each participant.

1) Model's Input Preparation: This subsection discusses the data preparation procedure for our FNN model. The multimodal dataset, which includes speech, pupillary, and visual features, is methodically preprocessed to ensure compliance with the model's input specifications. This preparation involves smoothing and alignment techniques, feature scaling, and organization into the appropriate tensor structure for efficient training and inference.

a) Visual features: The visual features, represented as X_v, are derived from raw eye-tracking data and typically contain high-frequency noise due to recording artifacts. To address this, a smoothing f_{fil} is applied to reduce noise while preserving meaningful patterns. The smoothed visual features denoted as $\tilde{X}_v$, are obtained using an Exponential Moving Average filter with a smoothing factor α set within the range $0.1 \leq \alpha \leq 0.2$, ensuring a balance between noise reduction:

$$\tilde{X}_{v,t} = \alpha X_{v,t} + (1 - \alpha)\tilde{X}_{v,t-1}$$

Here, $\tilde{X}_{v,t}$ represents the smoothed value at time t, $X_{v,t}$ is the raw data point at time t, and α (set empirically) is the smoothing factor controlling the degree of noise reduction. This step outputs quality-controlled visual features $\tilde{X}_v$ for further analysis.

b) Speech feature: The speech features, denoted as X_s, undergo a sequence of preprocessing operations to ensure consistency across recordings. Initially, the data is normalized and cleaned to remove unwanted noise. Afterward, the features are temporally aligned using a timed text grid alignment function, f_{alg}, which synchronizes the speech data with other modalities. The aligned speech features, $\tilde{X}_s$, are calculated as:

$$\tilde{X}_s(t) = X_s(k) + \frac{X_s(k+1) - X_s(k)}{t_{k+1} - t_k} \cdot (t - t_k).$$

In this equation, $X_s(k)$ and $X_s(k+1)$ are the speech data values at timestamps t_k and t_{k+1}, respectively, and t is the target timestamp. This linear interpolation process ensures the features are aligned to a shared temporal scale, facilitating seamless integration with other modalities.

c) Pupillary features: The pupillary features, X_p, are baseline-adjusted to account for physiological differences across participants. A baseline value, B_p, is computed as the mean pupil size over a specified time interval T:

$$B_p = \frac{1}{T} \sum_{t=1}^{T} X_p(t).$$

The baseline-adjusted features, $\tilde{X}_p$, are obtained by subtracting this baseline from the raw data:

$$\tilde{X}_p = X_p - B_p.$$

This adjustment standardizes the pupillary features, ensuring consistency across different individuals and reducing inter-participant variability.

2) NN Architecture: The proposed architecture consists of three distinct neural networks, each dedicated to processing a specific modality: visual, speech, and pupillary features. These networks extract modality-specific features and combine them to make predictions.

For each modality, a feedforward neural network processes the preprocessed feature set. The visual features $\tilde{X}_v$, speech features $\tilde{X}_s$, and pupillary features $\tilde{X}_p$ are passed through their respective neural networks, denoted as f_v, f_s, and f_p. The outputs of these modality-specific networks are represented as:

$$h_v = f_v(\tilde{X}_v), \quad h_s = f_s(\tilde{X}_s), \quad h_p = f_p(\tilde{X}_p),$$

TABLE III
RESULTS FROM THE CLASSIFICATION ANALYSIS USING FUSION NEURAL NETWORK. BOLD VALUES INDICATE SIGNIFICANT IMPROVEMENTS ($p<0.05$)

Task	Group	Modality	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC (%)	p-value (ACC)
Reading Task	Control vs. MCI	Gaze	71.5	69.3	70.8	69.5	0.682 ± 0.021	0.035
		Speech	52.2	51.4	51.1	50.5	0.495 ± 0.011	0.012
		Gaze + Speech	83.7	82.4	82.8	81.9	**0.791 ± 0.022**	-
	Control vs. AD	Gaze	73.1	76.5	71.7	75.4	0.702 ± 0.025	0.038
		Speech	53.4	55.2	51.9	51.3	0.514 ± 0.016	0.017
		Gaze + Speech	86.2	89.1	82.6	88.3	**0.823 ± 0.031**	-
	Control vs. MCI vs. AD	Gaze	64.3	65.8	62.5	63.2	0.613 ± 0.024	0.040
		Speech	47.5	48.3	46.8	45.9	0.458 ± 0.022	0.028
		Gaze + Speech	75.9	77.2	73.8	75.4	**0.725 ± 0.021**	-
King-Devick Test	Control vs. MCI	Gaze	64.7	62.4	63.5	63.1	0.612 ± 0.017	0.030
		Speech	47.8	45.5	47.1	44.7	0.445 ± 0.018	0.019
		Gaze + Speech	75.2	73.6	76.4	74.8	**0.728 ± 0.012**	-
	Control vs. AD	Gaze	68.4	69.2	74.6	72.3	0.671 ± 0.015	0.035
		Speech	50.3	50.9	49.5	48.4	0.492 ± 0.019	0.015
		Gaze + Speech	80.7	81.5	88.9	85.3	**0.798 ± 0.014**	-
	Control vs. MCI vs. AD	Gaze	58.6	55.7	58.9	56.2	0.552 ± 0.029	0.033
		Speech	41.9	40.5	39.7	39.8	0.385 ± 0.021	0.034
		Gaze + Speech	67.4	65.9	63.6	66.5	**0.689 ± 0.028**	-
Combined Task	Control vs. MCI	Gaze	70.9	68.4	66.2	68.6	0.668 ± 0.020	0.030
		Speech	50.6	49.3	49.1	48.7	0.471 ± 0.012	0.023
		Gaze + Speech	81.5	80.3	79.9	78.8	**0.765 ± 0.023**	-
	Control vs. AD	Gaze	72.7	70.9	72.8	69.7	0.675 ± 0.022	0.039
		Speech	53.7	52.5	53.3	52.6	0.502 ± 0.015	0.029
		Gaze + Speech	85.9	82.8	85.4	82.7	**0.805 ± 0.020**	-
	Control vs. MCI vs. AD	Gaze	60.5	57.8	61.4	58.6	0.582 ± 0.018	0.021
		Speech	43.4	41.8	40.7	40.9	0.392 ± 0.017	0.017
		Gaze + Speech	70.3	67.9	72.7	71.4	**0.728 ± 0.022**	-

where h_v, h_s, and h_p are the feature representations for visual, speech, and pupillary modalities, respectively.

Each layer l in the modality-specific networks applies the transformation:

$$h_m^{(l)} = \sigma(W_m^l h_m^{(l-1)} + b_m^{(l)}), \quad m \in \{v, s, p\}.$$

Here, W_m^l and $b_m^{(l)}$ are the weight matrix and bias vector for layer l of modality m, σ is the activation function (ReLU), and $h_m^{(l-1)}$ is the output from the previous layer.

The h_v, h_s, and h_p are concatenated to form a single feature vector:

$$h_c = \gamma(h_v, h_s, h_p),$$

where γ is a concatenation process. This combined feature vector h_c aggregates information across all modalities, enabling the model to capture interactions between them effectively.

Finally, the concatenated feature vector is passed through a Softmax classifier to predict class probabilities. The output prediction $\hat{y}$ is given by:

$$\hat{y} = \eta(W_o h_c + b_o),$$

where W_o and b_o are the output layer's weight matrix and bias vector, η the Softmax function ensures that the output is a probability distribution over the possible classes.

3) Implementation and Evaluation Setup: We used 10-fold cross-validation because the dataset has a limited sample size. In this method, 80% of the data were utilized for feature selection and model development, with 20% of the data serving as the evaluation test set. We utilized the Adam optimizer for the hyperparameter configuration to minimize the loss function during training efficiently [40]. The model was trained with a batch size of 64 and used cross-entropy loss; the learning rate was dynamically adjusted based on validation loss, with a minimum value of $10e^{-7}$ determined by the Adam algorithm's initialization with an LR of $10e^{-3}$, chosen for its stability during hyperparameter tuning. We considered accuracy, recall, and F1-score as evaluation metrics.

III. RESULTS

In this section, we present the classification outcomes of the Fusion Neural Network. We further evaluated and discussed the key feature sets that significantly contributed to the FNN's classification accuracy by utilizing SHAP analysis. Additionally, we performed a complexity analysis of the reading test and compared the performance of the FNN with traditional machine learning models.

A. Classification Performance

Initially, we conducted binary classifications between different groups, specifically Control vs. MCI and Control vs. AD. Following this, we extended the analysis to multi-class classification involving Control, MCI, and AD. Table III presents a summary of the classification results. We classified using individual tasks, specifically focusing on the Reading and King-Devick tasks. As shown in Table III, for the Reading task, our model achieved an accuracy of 0.78 ± 0.02 and 0.82 ± 0.03 in the binary classifications between Control vs. MCI and Control vs. AD, respectively. Following this, we extended the analysis to a three-way classification among all participant groups, where the FNN attained a 75% accuracy, along with an AUC score of 0.72 ± 0.02.

On the KD tests, our neural network achieved its highest AUC value of 0.79 ± 0.01 in the binary classification between the Control and AD groups. Following this, we performed classification using combined tasks by merging the data from both the Reading and KD tasks to predict across all tasks, aiming to assess potential synergies between them. Interestingly, the

results indicated that the Reading task alone produced higher performance metrics than the KD and combined tests. This trend is evident in the binary and multiclass classifications, as shown in Table III. We conducted a series of analyses to check how FNN performs on unimodal and multimodal data. The experimental results indicate that multimodality yields a much better performance than multimodal.

Table III demonstrates that eye-tracking (gaze) achieves a classification accuracy of approximately 60-70% in reading tasks, while speech alone yields lower performance, achieving only 50-60%. When we combined gaze and speech data, the classification performance improved significantly, with an 18% increase compared to using gaze or speech data individually. A similar trend is observed in KD tests, where the FNN classifier showed a 14% improvement. In binary classification tasks using combined data, our FNN model achieved accuracies of 81% and 85% for distinguishing control/MCI and control/AD, respectively. However, in the more complex three-way classification task, the performance dropped by 5-7%, reflecting the increased difficulty of distinguishing between classes. Additionally, the p-values of ACC reported in Table III highlight a strong correlation between unimodal and multimodal data, with multimodal approaches outperforming unimodal ones. Specifically, multimodality demonstrated a 15-23% improvement in both binary and three-way classification tasks. Furthermore, gaze data exhibited a significant correlation with the combined multimodal gaze + speech data, with p-values ($p < 0.05$) indicating statistically superior performance compared to speech data alone.

To validate the effectiveness of the proposed FNN model, we compared its performance with traditional ML methods. As shown in Table IV, the FNN significantly outperformed the ML methods and dense networks. For binary classification tasks between Control vs. MCI and Control vs. AD, the FNN achieved a mean accuracy improvement of 9.2%-13% compared to Support vector machines (SVM), Random Forest, and dense networks. Similarly, in the three-way multi-classification task (Control vs. MCI vs. AD), the FNN attained a mean accuracy of 70.6 ± 0.1, surpassing all other models under the same conditions. The p-values reported in Table are less than 0.001 for all classification tasks, underscoring the statistical significance of the FNN's superior performance over the competing methods.

Figure 3 illustrates the AUC scores for binary and multi-class classification tasks on combined (KD + Reading) tasks using multi- modal data. The proposed method achieved an AUC of 0.791 ± 0.02 for controls vs. MCI, 0.823 ± 0.03 for Control vs. AD, and 0.725 ± 0.01 for Control vs. MCI vs. AD. Moreover, for the KD task, the method achieved AUC of 0.728 ± 0.01, 0.798 ± 0.01, and 0.689 ± 0.02, shown in figure 4. On the combined tasks, the proposed method demonstrated slightly lower AUCs compared to the individual reading task, highlighting the challenges of aggregating data across modalities.

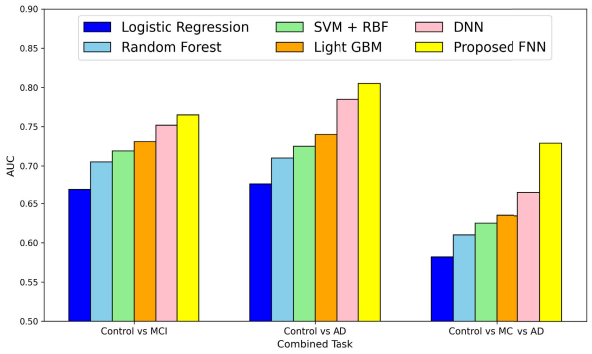
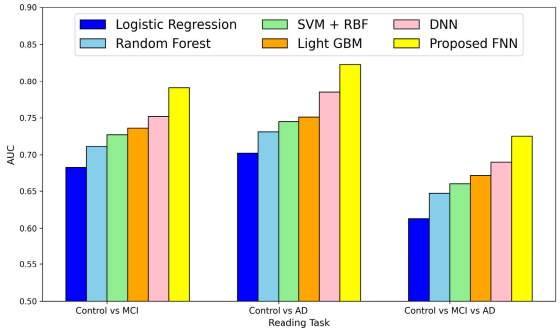
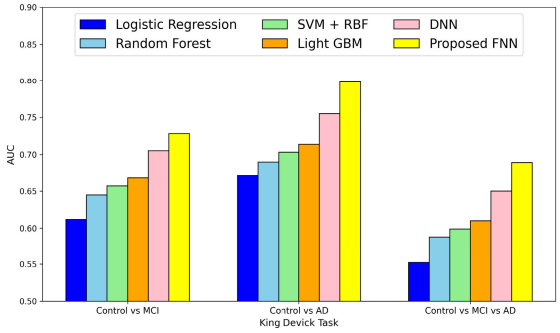


Fig. 3. AUC comparison between proposed FNN and other ML models on Combined (KD + Reading) task using multimodal data.



(a)



(b)

Fig. 4. Comparison of AUC scores between FNN and ML models. (a) shows the AUC comparison on reading task while (b) shows the KD test AUC scores.

B. Reading Task Complexity Analysis

Our results indicate that the reading task performed the highest classification accuracy in distinguishing between Control, MCI, and AD participants. To further analyze this performance, we conducted a complexity analysis to identify which words were correctly classified by the model for each participant group. For this evaluation, we categorized the words in the reading task into "easy" and "difficult" categories based on their Morphological, Lexical, and Syntactic Complexity, as defined by the Niilo Mäki Institute [34]. An example of such a rare word is *ilmastomääritelmän*, which roughly translates to the English genitive form of "a definition of climate" and may never have been spoken aloud by the participants.

Our analysis revealed that the model was more successful in correctly classifying difficult words such as *ilmastomääritelmän, syyspäiväntasaus, sukupuolellakin*. This is likely because these words provided richer information

TABLE IV
COMPARISON OF ABLATION RESULTS IN BINARY AND THREE-WAY CLASSIFICATION ON COMBINED MULTIMODAL DATA. BOLD FONT INDICATES THE BEST PERFORMANCE

Method	CO vs. MCI				CO vs. AD				CO vs. MCI vs. AD			
	ACC (%)	SEN (%)	SPE (%)	*p*-values (ACC)	ACC (%)	SEN (%)	SPE (%)	*p*-values (ACC)	ACC (%)	SEN (%)	SPE (%)	*p*-values (ACC)
Logistic Regression	67.2±1.5	65.4±2.1	63.2±2.2	<0.001	70.2±1.6	68.9±2.0	67.1±2.1	<0.001	59.5±1.3	57.5±1.9	60.5±2.4	<0.001
SVM + RBF	71.1±1.6	69.3±2.0	67.3±2.4	<0.001	74.7±1.7	72.8±2.3	70.9±2.5	<0.001	63.5±1.5	60.5±2.1	64.5±2.3	<0.001
Random Forest	72.7±1.4	71.0±2.1	68.1±2.3	<0.001	76.4±1.5	74.6±2.2	72.2±2.4	<0.001	65.0±1.6	62.5±1.9	66.0±2.1	<0.001
Light GBM	74.3±1.3	72.5±2.0	69.8±2.2	<0.001	78.2±1.4	76.1±2.1	74.2±2.3	<0.001	66.0±1.4	63.2±1.8	67.5±2.0	<0.001
Neural Network	75.3±1.4	73.7±1.8	70.8±2.1	<0.001	79.5±1.3	77.4±2.0	75.2±2.2	<0.001	67.0±1.3	64.5±1.7	68.2±2.0	<0.001
Proposed FNN	**79.1±0.2**	**77.3±0.3**	**74.2±0.3**	-	**83.4±0.2**	**82.7±0.1**	**85.1±0.2**	-	**70.6±0.1**	**67.6±0.3**	**72.3±0.1**	-

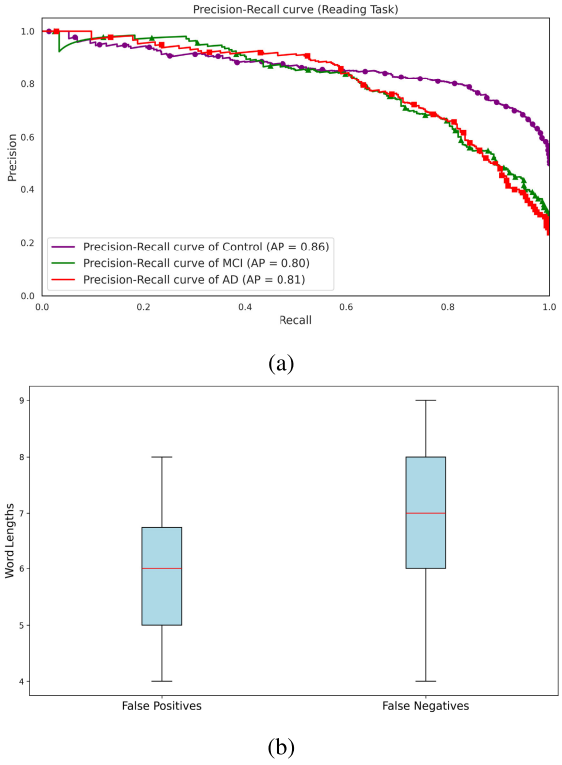


Fig. 5. (a) shows the precision-recall curve of FNN using multimodal gaze + speech data on reading task. (b) illustrates the box plot of average length of misclassified words by FNN in reading task.

about the participants' cognitive states. For instance, individuals with AD tended to read these difficult words with greater concentration, leading to distinctive eye movement patterns compared to those with MCI or controls. These differences made these ROIs valuable for focused analysis. On the other hand, the model struggled with easy ROIs, likely because shorter words or simpler eye movements are common across all participant groups. This similarity in eye movement patterns among different groups likely contributed to the model's difficulty in accurately classifying these easier ROIs.

Figure 5a presents Precision-Recall curves for the reading task, indicating that the model achieves the highest discriminative capability for the control group AP = 0.86, followed by comparable performance for MCI AP = 0.80 and AD AP = 0.81. While, in 5b, the model inaccurately classifies words of various lengths, with a slight preference for mid-length terms (4-8 letters). Similarly, in the case of false negatives, the mis-classifications cover a wide range of word lengths, with a notable concentration around mid-length words (4-9 letters). These distributions indicate that the algorithm struggles to identify the ROIs that are neither short nor too lengthy. This is likely because short words are fairly common among participants, resulting in similar eye movement patterns across all groups, making it challenging for the model to distinguish between individuals.

C. Feature Importance Analysis

Interpretability is essential when a model's ability to generate predictions could prove equally crucial as the prediction's accuracy. The correct interpretation of a model's output improves the predictability and credibility of the model. Following classifications, we implemented SHAP to interpret our Fusion Neural Network and identify the features contributing most to its decisions.

Notably, pupil sizes dominated as the most influential factor, ranking in the top four before ROI features or the first multimodal speech feature, which indicates the time between the last fixation on an ROI and the corresponding speech utterance. The analysis also reveals that speech features had a significant impact on the model's decisions, particularly the delay from the first fixation on an ROI to the start of speaking and the delay from the last fixation on an ROI to the end of speaking. Additionally, eye movement features played a moderate role, with the number of saccades starting or ending on an ROI and the number of regressive saccades (gaze direction left) being particularly influential. From the analysis in Figure 6, we observed that FNN struggles to identify short saccades from the fact that these saccades tend to exhibit similar behavior across all groups, leading to confusion in the model's ability to make accurate predictions.

Interestingly, our analysis revealed that regressive saccades where the gaze moves in the opposite direction also played a significant role in the model's decisions. This may be due to the observation that individuals in the control group have better control over regressive saccades compared to those with MCI and AD, making it a distinguishing factor in the model's classification process. Figure 7 also highlights that speech features exhibit a crucial role in the model's decision-making, particularly the delay in seconds from the first fixation on an ROI to the beginning of saying the ROI and the delay from the last fixation on an ROI to finishing saying the ROI. Additionally, eye movement features showed a moderate

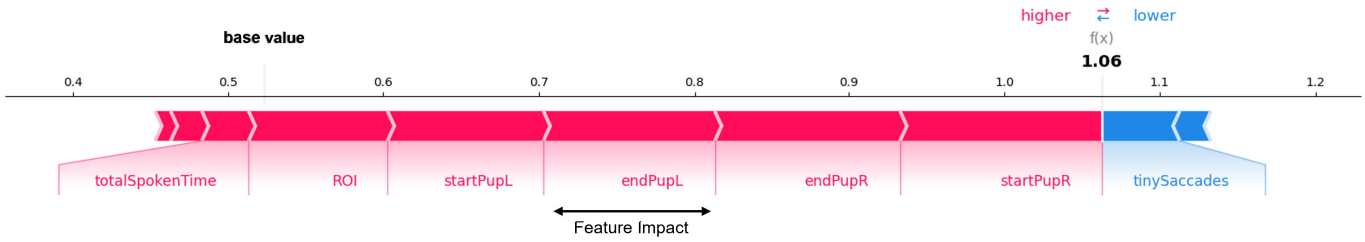


Fig. 6. Illustrates Top-ranked features from the Fusion network across multimodal data. It shows how individual features influence the model's prediction. Red features, such as pupil size and total spoken time, strongly increase the output, while the blue feature, tinySaccades, slightly decreases it.

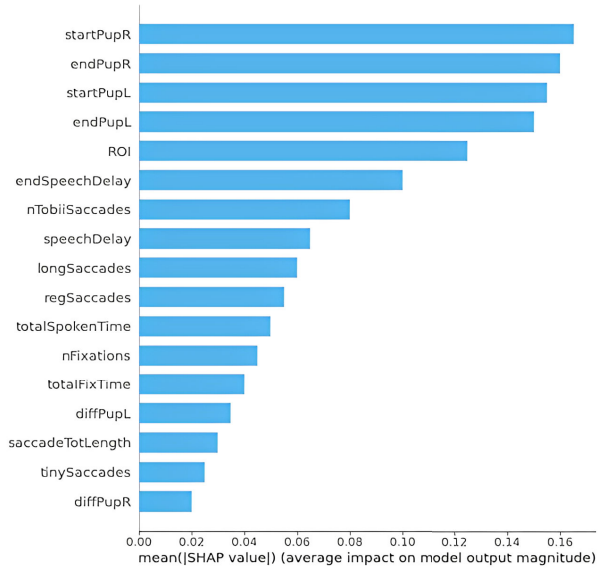


Fig. 7. Feature importance of the ROI-based analysis obtained by FNN. Pupil sizes seem to be the most important factor with speech features.

impact, with the number of saccades starting or ending on an ROI and the number of regressive saccades (gaze direction left) being notably influential. Overall, pupil-based features, when combined with speech features, were consistently ranked as more important than other visual features in both the Reading and KD tasks.

D. Behavioral Features

1) Statistical Analysis: Following the approach of Hannonen et al. [10], we conducted a one-way analysis of variance (ANOVA) on outcome measures (e.g., total time, No of errors) across the three groups (CO, MCI, AD). When the ANOVA indicated an overall difference, we performed least significant difference (LSD) post-hoc comparisons to determine which specific pairwise contrasts (CO–MCI, MCI–AD, CO–AD) were statistically significant ($p<0.05$). This method allowed us to pinpoint the exact group differences in both the reading task and KD results. Our analysis of behavioral features focuses on the reading task and KD test. Both tasks provide important information about cognitive processing speed and error propensity among CO, MCI, and AD. Table V provides a distribution of behavioral data, such as the error numbers and total time for each group. In our reading task, the mean total duration for controls was 88.89 ± 18.16s, compared to 90.01 ± 12.71s for the MCI group and 99.85 ± 24.22s for the AD group. Although the trend indicates longer task durations with increasing disease severity, pairwise comparisons did not reveal statistically significant differences between CO and MCI ($p = 0.821$) or between MCI and AD ($p = 0.154$); the difference between controls and AD approached significance ($p = 0.090$). Word error rates in the reading task were more revealing. Controls made 2.28 ± 2.82 errors on average, while MCI subjects committed 1.58 ± 2.06 errors. In contrast, AD participants exhibited a significant increase in errors (4.73 ± 5.11). Statistical analysis confirmed significant differences in word errors when comparing AD with both controls ($p = 0.039$) and MCI ($p = 0.026$), whereas the difference between controls and MCI did not reach significance ($p = 0.377$).

In KD, the test results revealed clear group differences. Controls demonstrated the fastest reading performance (mean total time: 69.3 ± 14.2 s), whereas AD patients took significantly longer (82.7 ± 16.5 s). The MCI group (76.6 ± 14.2 s) occupied an intermediate position; however, while the difference between controls and AD reached statistical significance ($p = 0.008$), the differences between controls and MCI or between MCI and AD were not statistically significant. In addition, the error analysis further differentiated the groups: AD participants committed more errors (2.74 ± 3.62) compared with controls (0.62 ± 1.81) and MCI individuals (0.18 ± 0.53), with pairwise comparisons showing significance between CO and AD ($p = 0.005$) and between MCI and AD ($p = 0.004$).

Our findings show that AD exhibits reduced reading speed and a higher error rate relative to both controls and MCI subjects. The KD test provides robust differentiation in both timing and error measures, particularly between the AD and control groups. The reading task shows a similar pattern in error rates, with AD subjects performing significantly worse than the other groups.

IV. Discussion

In this study, we analyzed a multimodal dataset that includes two different tasks for MCI/AD classification. This study expands on a journal paper that our group previously published [10], highlighting the potential of eye movement data obtained from the King-Devick task. They utilized statistical analysis on a few simple eye-tracking metrics to show that abnormal gaze metrics, especially differences in saccadic amplitude and their temporal duration, and were able to show significant group-wise differences between healthy controls

TABLE V
DISTRIBUTION OF BEHAVIORAL DATA FOR EACH GROUP IN READING TASK AND KD TEST

Task	Measure	CO (mean ± SD)	MCI (mean ± SD)	AD (mean ± SD)	CO–MCI (*p*)	MCI–AD (*p*)	CO–AD (*p*)
Reading Task	Total time (s)	88.9 ± 18.1	90.1 ± 12.71	99.8 ± 24.22	0.821	0.154	0.090
	Number of errors	2.28 ± 2.82	1.58 ± 2.06	4.73 ± 5.11	0.377	0.026	0.039
King-Devick Test	Total time (s)	69.3 ± 14.2	76.6 ± 14.2	82.7 ± 16.5	0.319	0.665	0.008
	Number of errors	0.62 ± 1.81	0.18 ± 0.53	2.74 ± 3.62	1.000	0.004	0.005

and persons with MCI with an AUC of 0.64 for either feature. Extending on our prior research, we added a reading task to investigate the two data modalities (eye movements and speech features) in more detail for classifying MCI/AD compared to controls.

Referring to the results of this paper, multimodal data (gaze + speech) demonstrates a significant improvement in classification accuracy compared to unimodal data (speech or gaze alone). The superiority of multimodal data can be attributed to its ability to integrate complementary information from different modalities, which enhances the model's capacity to capture diverse patterns in participants' behavior. Our experiments from table III confirmed that combining gaze and speech data allows the model to leverage both the temporal dynamics of speech and the spatial characteristics of gaze, resulting in better feature representation and improved classification performance

The results indicate that the reading task outperforms the King-Devick test in providing rich information for the final classification. This is likely because the reading task involves a more complex interaction of cognitive processes. Prior studies have shown that reading engages multiple cognitive domains, including language processing, working memory, and executive function, all of which are affected in MCI and AD [41]. Compared to the rapid number-naming KD test, the reading task provides a higher degree of linguistic and syntactic complexity, which helps differentiate between Control, MCI, and AD participants more distinctly [42].

One key challenge with DL algorithms in medical diagnostics is their lack of interpretability. Most ML/DL algorithms operate as black boxes, providing predictions without clear explanations. To address this limitation, we employed SHAP analysis to interpret the model's decision-making process and assess its accuracy.

Our analysis revealed that the most influential features of the classifier included pupil size and speech-derived metrics. Pupil sizes were baselined per participant during preprocessing and normalized across the dataset to minimize the potential for unintended interference. Interestingly, despite the well-documented challenges in accurately estimating pupil size [43], these features demonstrated strong relevance to the classifier's performance. This unexpected finding suggests the potential of pupil size as a biomarker, warranting further investigation using consistent experimental conditions.

A. Limitations

We noticed a few challenges during the experiment. One key issue was hyperparameter tuning of our proposed FNN, in which each subnetwork processes different types of data. Selecting a single set of optimal hyperparameters was difficult and required trial and error. We also noticed that the speech subnetwork consistently underperformed then eye movement and pupil subnetworks. This is due to the low volume of speech in some samples, making it harder for the model to learn meaningful patterns from the audio. As a result, the fusion layer sometimes relied less on the speech features. To address these issues, we plan to apply better speech pre-processing techniques in future work. Another shortcoming of our study is the relatively small size of the dataset. Although our dataset aligns with those utilized in similar research efforts [6], [14], [22], [31], [44] and has enabled us to achieve strong classification performance, its size constrains the application of more advanced and robust machine learning techniques. To address this, we plan to expand the dataset to at least 500 participants, ensuring equal distribution across groups. This larger and more balanced data set will improve the generalizability of our findings and allow the adoption of more sophisticated methodologies for deeper insights and improved predictive accuracy.

V. CONCLUSION AND FUTURE WORK

Early prediction of MCI and AD is critical, as researchers believe that early diagnosis will be key to slowing or even stopping the disease. In this study, we addressed the challenge of predicting MCI and AD from healthy controls using a multimodal dataset that includes eye-tracking and speech data collected during the King-Devick test and a reading task. Building on previous work based on statistical analysis, we applied a Fusion Neural Network model to integrate eye movement samples, speech features, and pupillary features, aiming to classify MCI/AD patients from healthy controls. Our fusion classifier achieved an accuracy of 79.2% for MCI diagnosis and 82% for AD. To enhance the interpretability of our deep learning model, we implemented SHAP analysis, which provides an explainable AI approach by revealing the contributions of different features to the model's decisions. Our findings highlight that features related to pupil size and eye-speech temporal dynamics are particularly strong indicators for these detection tasks. Recruitment for this study is ongoing, and the cohort size continues to grow. Participants have also agreed to complete follow-up assessments every six months over two years, enabling us to track disease progression over time. In future assessments, we will focus on technical improvements to our deep learning models, particularly by integrating an interactive large language model to leverage eye-tracking and language features more effectively.

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